

Institute of Actuaries of India

**Subject ST6 – Finance and
Investment B**

October 2014 Examination

INDICATIVE SOLUTION

Solution 1:**i.)**

The capital requirement is calculated by a equity market stress. The put options will increase in the value when performing the equity market fall stress which would compensate the overall reduction in assets.

The key risks are

1. Counterparty risk
2. Basis mismatch risk assets may not be similar to NSE.

[3]**ii.) & iii.)**

	Option 1	Option 2	Option 3
Strike	5500	5500	6000
Number of option	20000	20000	20000
Time to expiry	6	9	6
Dividend yield	5%	5%	5%
Current Market	8000	8000	8000
1b	Current value of options	Current value of options	Current value of options
Forward 6 year	currentval * exp (intrate- divid) * time	currentval * exp (intrate- divid) * time	currentval * exp (intrate- divid) * time
Forward 6 year	8,753	10,480	8,753
Money ness	63%	52%	69%
Volatility	25%	32%	26%
interest rate	6.50%	8.00%	6.50%
Output:			
D1	107%	115%	92%
D2	45%	19%	29%
N(D1)	0.85657	0.87525	0.82042
N(D2)	0.67460	0.57595	0.61499
Call Price	2,564.40153	2,922.75255	2,363.96015
Put Price	361.67	498.86	499.76
Value of Put Options (000)	7,233,371.57	9,977,294.88	9,995,112.59

The table above calculate the value of option for section a and b

[4+8=12]

iv.)

	Option 2	Option 3
Strike	5500	6000
Number of option	20000	20000
Time to expiry	9	6
Divident yield	5%	5%
Current Market	4000	4000
1b	Current value of options	Current value of options
	$\text{currentval} * \exp(\text{intrate} - \text{divid}) * \text{time}$	$\text{currentval} * \exp(\text{intrate} - \text{divid}) * \text{time}$
Forward 6 year	5,240	4,377
Money ness	105%	137%
Volatility	27%	17%
interest rate	8.00%	6.50%
Output:		
D1	34%	-55%
D2	-46%	-97%
N(D1)	0.63177	0.29138
N(D2)	0.32332	0.16708
Call Price	745.78468	184.70490
Put Price	872.41	1,283.77
Value of Put Options (000)	17,448,189.54	25,675,465.30

Key assumptions are same as BS no change in volatility surface and dividend yields.

[8]

v.)

It is better to use option 3 given it provide higher capital saving.

Other costs

- 1) Transaction cost of selling first option and purchasing other options
- 2) Timing of transaction is crucial as it may impact the profit captured.
- 3) First option may be more useful if company is looking at forward projections.

[3]

[26 Marks]

Solution 2:

i.) The value of annual payment given by the following equation

$$X = \max(A, \min(Q(n), B))$$

$Q(n)$ = accumulation of RPI index

$$B = (1.06)^{15}$$

If $Q(n)$ is less than A the victim effectively has a European put option which gives him the right to sell (at the vesting date i.e. time 15) his payout of $Q(n)$ per annum for an annuity of A per annum.

If $Q(n)$ is greater than the upper limit of $B = R(n) = 1.06^{15}$, then the member has written a European call option for the company. The company has the right to buy the victim's payout of $Q(n)$ per annum with an annuity of $R(n)$ per annum.

[6]

ii.) Uses Black's formula for the above two options

$$V_A = Ae^{-\delta T} \Phi(f_1) + Q_0 e^{-\eta T} \Phi(f_2)$$

$$V_B = Q_0 e^{-\eta T} \Phi(d_1) - Be^{-\delta T} \Phi(f_2)$$

[6]

[12 Marks]

Solution 3:

i. Monte Carlo method uses a set of random numbers to project the path of economic variable using a suggested probability distribution. The projection is done for multiple simulations which provide a strong basis for representing the probable randomness of the variable itself.

[3]

ii. **Advantages:**

- a. Possibility of allowing multiple factors
- b. Successful in valuing many path dependent option
- c. Better scalability compared to tree structure
- d. Simple to understand though a brute force approach

Disadvantages:

- a. Need large scale computer program to work
- b. Slow convergence ($1/n$)
- c. Difficult to estimate delta and gamma
- d. Hard to get quasi random number sequence
- e. American options are tricky to value
- f. Intermediate paths are difficult to calculate (may lead to nested stochastic)

[6]

iii. Approach to review the Monte Carlo

- a. Antithetic variable
- b. Moment matching
- c. Stratified sampling
- d. Control variants
- e. Quasi Random sequence

[6]

[15 Marks]

Solution 4:

From A's point of view, by entering in to a fixed rate loan and entering into a swap with the bank, it effectively pays a floating rate of $(\text{LIBOR} - S'_A - F'_A) \%$. It is better off if $S_A > -S'_A - F'_A$.

Similarly from B's point of view, by issuing a floating rate loan and entering in to a swap, it effectively pays a fixed rate of $(F_B + F'_B + S_B + S'_B) \%$. It is better off if $F_B > F_B + F'_B + S_B + S'_B$.

Finally the bank is better off by entering in to both swaps if $F_B + F'_B - S'_A - F'_A - F'_A + S'_B > 0$.

Put these inequalities together and rearrange to get the final result.

[6 Marks]

Solution 5:

The price is given by:

$$\text{Price} = \frac{1}{r^{10}} \left[E^* \frac{S_1 + S_2 + \dots + S_{10}}{10} \right] = \frac{1}{10r^{10}} [E^*(S_1) + E^*(S_2) + \dots + E^*(S_{10})]$$

Now, note that $E^*[S_t] = S_0 r^t$ because in the risk-neutral probabilities, the stock price grows at the risk free rate.

Thus, the price becomes

$$\text{Price} = \frac{1}{10r^{10}} [(S_0 r) + (S_0 r^2) + \dots + (S_0 r^{10})] = \frac{S_0}{10r^{10}} [r + r^2 + \dots + r^{10}]$$

[7 Marks]

Solution 6:

$$\begin{aligned} & E \left[\frac{W(t)^2}{2} \mid W(u), 0 \leq u \leq s \right] \\ &= E \left[\frac{W(s)^2}{2} \mid W(u), 0 \leq u \leq s \right] + E \left[\frac{W(t)^2}{2} - \frac{W(s)^2}{2} \mid W(u), 0 \leq u \leq s \right] \\ &= \frac{W(s)^2}{2} + \frac{t-s}{2} \end{aligned}$$

Thus it is not a martingale.

[5 Marks]

Solution 7:

- i. Here, $r = 1.25$; $u = 2$ and $d = 0.5$. Thus

$$\text{Risk neutral probability, } \pi_u = \frac{r - d}{u - d} = \frac{1.25 - 0.5}{2 - 0.5} = \frac{r - 0.75}{1.5} = 0.5.$$

[2]

- ii. European put values:

Based on the evolution of the tree, the values at the final node are 0, 0 and 50.

Values at the nodes in period 1 are 0 and 20 respectively.

Value at the starting node is thus 8.

The stock price tree is given as

				400
		200		
100				100
		50		
				25
t = 0		t = 1		t = 2

Thus, the values at the node for the European option is given by

				0
		0		
8				0
		20		
				50
t = 0		t = 1		t = 2

[3]

- iii. American put values:

				0
		0		
10				0
		25		
				50
t = 0		t = 1		t = 2

[3]

- iv. Price of the European Call, using the Put – Call parity equation, is given by

$$c + PV(X) = S_0 + p \Rightarrow C = S_0 + p - PV(X)$$

Substituting all the numbers, we get

$$c = 100 + 8 - (75/(1.25^2)) = 100 + 8 - 48 = 60.$$

[2]

[10 Marks]

Solution 8:

The accumulated cost of the hedge is $(84.30 - 74.80)\exp(0.06) = 10.09$.

Let x be the market price. If $x < 0.12$ the put is in the money and the payoff is $10,000(0.12 - x) = 1,200 - 10,000x$.

The sale of the burger buns, thus have a payoff of $10,000x - 1,000$ for a profit of $1,200 - 10,000x + 10,000x - 1,000 - 10.09 = 190$.

From 0.12 to 0.14 neither of the options have a payoff and the profit is $10,000x - 1,000 - 10.09 = 10,000x - 1,010$.

If $x > 0.14$ the call is in the money and the payoff is $-10,000(x - 0.14) = 1,400 - 10,000x$. The profit is $1,400 - 10,000x + 10,000x - 1,000 - 10.09 = 390$.

The range, therefore, is 190 to 390.

[6 Marks]

Solution 9:**i. The Vasicek model and its statistical properties:**

- This is a model used for modelling the short rate of interest $r(t)$.
- It assumes that $r(t)$ has the dynamics of an Ito process (in fact, an Ornstein-Uhlenbeck process) under the risk-neutral probability measure.
- It is a one-factor model because it incorporates a single source of randomness.
- It assumes the model $dr(t) = \alpha[\mu - r(t)]dt + \sigma dW(t)$, where $W(t)$ is standard Brownian motion
- The movements in the interest rate are therefore normally distributed (and hence can take negative values).
- The parameter σ determines the volatility.
- The parameter α is assigned a positive value, so that $r(t)$ is stationary and mean reverts to the constant value μ . A higher value of α implies more rapid mean reversion.

[3]

ii.**a. Condition for a stationary point**

We can rewrite the given function as:

$$f(t) = r_0 e^{-\alpha t} + r_\infty (1 - e^{-\alpha t}) + k(1 - e^{-\alpha t})e^{-\alpha t} = r_\infty + (r_0 - r_\infty + k)e^{-\alpha t} - k e^{-2\alpha t}$$

Differentiating,

$$f'(\tau) = -(r_0 - r_\infty + k)\alpha e^{-\alpha t} + 2k\alpha e^{-2\alpha t} = [-(r_0 - r_\infty + k) + 2k e^{-\alpha t}]\alpha e^{-\alpha t}$$

For a stationary point, this must equal zero and so the expression in square brackets must also equal zero. Thus

$$-(r_0 - r_\infty + k) + 2k e^{-\alpha t} = 0 \Rightarrow e^{-\alpha t} = \frac{r_0 - r_\infty + k}{2k}$$

b. Shapes of graphs

Graph (1) will have a stationary point if

$$e^{-\alpha\tau} = \frac{0.04 - 0.06 + 0.1}{2(0.1)} = 0.4$$

There will be one solution to this equation, corresponding to a positive value of τ .

Also when $\tau = 0$, the slope is

$$f'(0) = [-(r_0 - r_\infty + k) + 2k]\alpha = [k - (r_0 - r_\infty)]\alpha = [0.1 - (0.04 - 0.06)]\alpha > 0$$

So graph (1) will be a humped graph starting at 0.04 (4%) for the short rate and ending at 0.06 (6%) for the long rate.

Graph (2) will have the stationary point if

$$e^{-\alpha\tau} = \frac{0.06 - 0.04 + 0.02}{2(0.02)} = 1$$

So this will have a stationary point at $\tau = 0$ only.

So Graph (2) will start at 0.06 (6%) and then fall steadily to 0.04 (4%).

c. When might the yield curve like Graph 1 occur?

The usual explanation for a humped yield curve is market segmentation. Different categories of investors are attracted to bonds with different terms (short, medium or long). The bond prices, and hence the yields within each “segment” of the market, are determined by demand and supply.

In fact, graph (1) is a common shape for the yield curve.

[6]

iii. Advantages and limitations

- The mean-reverting property is consistent with real-world observations.
- The model is mathematically tractable, with few parameters to estimate.
- The statistical properties are based on the familiar normal distribution.
- A variety of shapes can be obtained for the yield curve, *i.e.* upward-sloping, downward sloping and humped.
- The model leads to arbitrage-free yield curves.
- The model has the disadvantage that it allows $r(t)$ to take negative values, which we would not expect to observe in the real world.

[4]

[13 Marks]
